

HARSHITHA OBILANENI

Data Engineer

Pleasant Hill, CA | +1 (361) 466-7481 | obilaneni@gmail.com | [Linkedin](#)

SUMMARY

Data Engineer with 3+ years of experience building scalable ETL/ELT pipelines and cloud-based data platforms across healthcare and financial domains. Skilled in PySpark, Azure Databricks, Delta Lake, and Hadoop, delivering solutions that reduced processing time by 40%, lowered data errors by 60%, and generated \$2M+ in savings. Experienced in cloud migration, CI/CD automation, governance, real-time analytics, and ML enablement.

EXPERIENCE

Data Engineer | Optum, USA

May 2024 – Present

- Engineered scalable PySpark ETL pipelines processing 10TB+ healthcare claims data daily, reducing processing time by 40% and enabling near real-time analytics for 5M+ patients.
- Optimized Azure Databricks workflows using Delta Lake (ACID transactions), cutting data latency from hours to minutes and improving downstream ML query performance by 3x.
- Built REST API and microservices-based ingestion frameworks, automating 500K+ records/hour into Azure Data Lake while ensuring HIPAA compliance, supporting fraud detection initiatives that saved \$2M annually.
- Led code reviews, CI/CD deployments, and production releases for analytics applications, resolving 95% of incidents within SLA and increasing team velocity by 25%.
- Partnered with data scientists to design and prototype feature stores, accelerating model training cycles by 50% and enabling personalized care recommendations for high-risk cohorts.

Data Engineer | Tata Consultancy Services (TCS), India

Jun 2022 – Jul 2023

- Developed Spark-Hive ETL pipelines on Hadoop processing 2TB+ datasets, reducing batch runtime by 35% and delivering timely BI dashboards for enterprise reporting.
- Implemented data quality frameworks using SQL-based validations and monitoring, decreasing data errors by 60% across 10+ pipelines and preventing \$500K in rework for banking clients.
- Led migration of legacy on-prem systems to Azure Data Lake, processing 1M+ transactions daily with zero downtime, enhancing scalability for cloud-first initiatives.
- Troubleshoot and optimized unstable pipelines, achieving 98% system uptime and mentoring junior engineers to reduce incident resolution time by 50%.
- Designed modular, scalable data models supporting multi-source ingestion (SQL/NoSQL), enabling agile delivery of 15+ client features ahead of schedule.

SKILLS

- Data Engineering & Big Data:** PySpark, Apache Spark, Spark-Hive, Hadoop Ecosystem, ETL/ELT Pipelines, Delta Lake (ACID, Partitioning, Z-Ordering), Data Lakes/Warehouses
- Cloud Platforms:** Azure Databricks, Azure Data Lake, Azure DevOps, GCP (exposure)
- Programming & Querying:** Python, SQL, Data Modeling (Modular/Reusable), REST APIs, Microservices Integration
- DevOps & Automation:** CI/CD Pipelines, Git, Automated Monitoring, Performance Tuning (Caching, Broadcast Joins)
- Data Quality & Governance:** SQL Validations, Reconciliation Checks, HIPAA Compliance, Incident Management, Root Cause Analysis
- Methodologies:** Agile/Scrum, Feature Stores, ML-Ready Datasets, Streaming Pipelines, Zero-Downtime Migrations

EDUCATION

Master of Science in Computer Science

Aug 2023 – May 2025

Texas A&M University–Kingsville, USA

Bachelor of Technology in Electronics & Communication Engineering

Aug 2018 – May 2022

Sasi Institute of Technology and Engineering, India

PROJECTS

Healthcare Claims Analytics Platform

- Engineered PySpark ETL pipelines on Azure Databricks processing 10TB+ daily claims for 5M+ patients.
- Delivered Delta Lake optimizations for 3x query speed and \$2M fraud savings via CI/CD automation.

Banking Data Warehouse Migration

- Led zero-downtime migration to Azure Data Lake with 1M+ transactions and 2TB datasets.
- Optimized Spark-Hive ETL for 35% faster runtimes, 60% error reduction, and Agile BI dashboards.

CERTIFICATIONS

- AWS Certified Machine Learning – Specialty
- Generative AI with Large Language Models – DeepLearning.AI